COHOMOLOGY OF A DERIVED LIMIT FOR A SUITABLE INVERSE SYSTEM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument, the passage from the objects of B to the whole site being supplied by the covering hypothesis (1).

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Derived limit of an inverse system of objects of the derived category. Same chapter together with the chapter Derived Categories, the defining triangle of the derived limit and its long exact cohomology sequence. Supplies the object R lim K_n and the exact sequence used in the proof.
- Vanishing of cohomology of the cohomology sheaves on the objects of B and vanishing of R^1 lim. Same chapter, the hypotheses (2)(a) and (2)(b) of the statement, which are the conditions under which cohomology of limits is computed by limits of cohomologies. Supplies the two vanishing statements that make the limit exact.

Source reference. Original remark 0BKW: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 5201–5201. Navigation pointer invoked at proof line 5523; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0BKZ: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 5467–5467. Navigation pointer invoked at proof line 5524; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Source reference. Original lemma 08U5: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 5532–5532. Navigation pointer invoked at proof line 5532; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let (K_n) be an inverse system of objects of $D(\mathcal{O})$. Let $\mathcal{B} \subset \mathrm{Ob}(\mathcal{C})$ be a subset. Assume*

(1) *every object of $\mathcal{C}$ has a covering whose members are elements of $\mathcal{B}$,*
(2) *for all $U \in \mathcal{B}$ and all $q \in \mathbf{Z}$ we have*
   (a) *$H^p(U, H^q(K_n)) = 0$ for $p > 0$,*
   (b) *the inverse system $H^0(U, H^q(K_n))$ has vanishing $R^1 \lim$.*

Then $H^q(R \lim K_n) = \lim H^q(K_n)$ for $q \in \mathbf{Z}$.

Proof. Set $K = R \lim K_n$. We will use notation as in Remark 0BKW. Let $U \in \mathcal{B}$. By Lemma 0BKZ and (2)(a) we have $H^q(U, K_n) = H^0(U, H^q(K_n))$. Using that the functor $R\Gamma(U, -)$ commutes with derived limits we have

$$H^q(U, K) = H^q(R \lim R\Gamma(U, K_n)) = \lim H^0(U, H^q(K_n))$$

where the final equality follows from More on Algebra, Remark 08U5 and assumption (2)(b). Thus $H^q(U, K)$ is the inverse limit the sections of the sheaves $H^q(K_n)$ over U. Since $\lim H^q(K_n)$ is a sheaf we find using assumption (1) that $H^q(K)$, which is the sheafification of the presheaf $U \mapsto H^q(U, K)$, is equal to $\lim H^q(K_n)$. This proves the lemma. □